

Ex. 3.18 / Ch. 3 / P97 / Chloe / 2025.7.27

时: 艾伦菲斯特定理 $\frac{d}{dt}\langle Q \rangle = \frac{i}{\hbar}\langle [\hat{H}, \hat{Q}] \rangle + \langle \frac{\partial Q}{\partial t} \rangle$

$$(a) Q=1, \frac{d}{dt}\langle 1 \rangle = \frac{i}{\hbar}\langle [\hat{H}, 1] \rangle + \langle \frac{\partial 1}{\partial t} \rangle = 0$$

又 $\langle 1 \rangle = \int_{-\infty}^{+\infty} |\Psi(x, t)|^2 dx$, 这表明波函数归一化常数不随时间变化, 若初始波函数归一化, 则始终归一化, 概率守恒

$$(b) Q=\hat{H}, \frac{d}{dt}\langle H \rangle = \frac{i}{\hbar}\langle [\hat{H}, \hat{H}] \rangle + \langle \frac{\partial \hat{H}}{\partial t} \rangle = 0 + \langle \frac{\partial \hat{H}}{\partial t} \rangle$$

若 $\hat{H}$ 不显含时间, 则 $\frac{d}{dt}\langle H \rangle = 0$, 即 $V=V(x)$ 与 t 无关, 能量期望值不随时间变化, 能量守恒; 若 $V=V(x, t)$, 则 $\hat{H}$ 含时, $\frac{d}{dt}\langle H \rangle \neq 0$, 因系统与外界有能量交换, 故系统内能量不守恒, 这在宏观与微观者一致。

$$(c) Q=x, \frac{d}{dt}\langle x \rangle = \frac{i}{\hbar}\langle [\hat{H}, x] \rangle + \langle \frac{\partial x}{\partial t} \rangle = \frac{i}{\hbar}\langle [\frac{\hat{p}^2}{2m} + V(x), x] \rangle$$

$$= \frac{i}{\hbar}\langle [\frac{\hat{p}^2}{2m}, x] + [V(x), x] \rangle = \frac{i}{\hbar} \cdot \frac{1}{2m}\langle [\hat{p}^2, x] \rangle = \frac{i}{\hbar} \cdot \frac{1}{2m}\langle [\hat{p}^2, x] \rangle$$

$$x[\hat{p}^2, x] = [\hat{p}\hat{p}, x] = \hat{p}[\hat{p}, x] + [\hat{p}, x]\hat{p} = \hat{p}(-i\hbar) + (-i\hbar)\hat{p} = -2i\hbar\hat{p}$$

$$\text{故 } \frac{d}{dt}\langle x \rangle = \frac{\langle p \rangle}{m}, \text{ 在经典力学中, 速度 } v = \frac{dx}{dt} = \frac{p}{m}, \text{ 量子力学中, 由 } \frac{d\langle x \rangle}{dt} = \frac{\langle p \rangle}{m}$$

知, 尽管微观粒子具有波粒二象性, 但统计均值遵循经典运动学关系, 即量子力学在宏观尺度下与经典力学的一致性。

$$(d) Q=\hat{p}, [\hat{H}, \hat{p}] = [\frac{\hat{p}^2}{2m}, \hat{p}] + [V(x), \hat{p}] = 0 + [V(x), \hat{p}] =$$

$$[V(x), \hat{p}]\psi = V(x)\hat{p}\psi - \hat{p}(V(x)\psi) = -i\hbar V(x)\frac{\partial \psi}{\partial x} - [-i\hbar(\frac{\partial V}{\partial x}\psi + V(x)\frac{\partial \psi}{\partial x})]$$

$$= i\hbar\frac{\partial V}{\partial x}\psi, \text{ 故 } [V(x), \hat{p}] = i\hbar\frac{\partial V}{\partial x}$$

$$\text{故 } \frac{d}{dt}\langle \hat{p} \rangle = \frac{i}{\hbar}\langle i\hbar\frac{\partial V}{\partial x} \rangle + \langle \frac{\partial \hat{p}}{\partial t} \rangle = -\langle \frac{\partial V}{\partial x} \rangle, \text{ 经典力学中 } F = -\frac{\partial V}{\partial x}, \text{ 结果}$$

$\frac{d\langle \hat{p} \rangle}{dt} = \langle F \rangle$, 即微观粒子的平均动量变化率等于平均力, 继承了经典牛顿第二定律, 说明量子力学在均值演化中遵循经典运动学关系。